

This forum features game-practitioner perspectives on the interaction design process, techniques, and evaluation involved in creating playful experiences. We focus on how technology advancement, infrastructure, and constraints shape the player experience. — **Pejman Mirza-Babaei, Editor**

Shaping the Future of Games in Society:

Insights from a Dagstuhl Perspectives Workshop

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In March 2025, a landmark gathering of leading games research experts took place at Schloss Dagstuhl in Germany, for the Dagstuhl Perspectives Workshop: The Future of Games in Society (<https://bit.ly/3H3ckxz>). This workshop tackled a critical question: How can we tap into the immense potential of games to create a positive societal impact? With 4 billion people worldwide engaging in games and more than a trillion hours of gameplay annually, the stakes couldn't be higher.

Over three days, the workshop stimulated rich conversations. We explored using games as powerful vehicles and resources for education, social change, well-being, and more. We proactively identified and discussed major potential ethical challenges in game design and implementation. Crucially, participants championed inclusivity, with guidelines to develop games that resonate with diverse individuals across age, gender, culture, and socioeconomic backgrounds. We discussed new opportunities and important responsibilities that come with games' rising cultural influence.

A MEETING OF DIVERSE PERSPECTIVES

The seminar, led by the authors of this article, brought together researchers and industry professionals from disciplines spanning psychology, computer science, design, health, education, and others. This cross-disciplinary approach reflected the fundamental nature of games research and the application of games in modern culture.

Games have changed. Today, they are not just entertainment but rather a highly diverse and applicable tool that sparks change across society. Such a paradigm shift demands a thoughtful and strategic approach to maximize games' potential. We collectively noticed that, as games become increasingly sophisticated and pervasive, their influence extends into healthcare, education, scientific research, and social change initiatives. This expansion of games warrants cautious evaluation of ethical effects and responsible development practices. The workshop established guiding principles to allow everyone to engage with the potential of games, promoting inclusivity and accessibility—core values that have been part of digital gaming's DNA since its inception at the edge of mainstream technology and culture.

EIGHT GAME RESEARCH THEMES

The workshop brought into focus eight pivotal themes, each representing

a critical direction for the future of games in society.

Games for human flourishing.

Games are powerful instruments that can help people flourish. They contribute to mental health, emotional resilience, and social connectivity [1]. Games that use player psychology, such as adaptive gameplay and emotional intelligence, can foster personal growth and build empathy. Games can help address anxiety, depression, and social isolation, while offering therapeutic benefits that traditional methods may not provide. Crucially, the industry must focus on ethical game design, steering away from manipulative parts, such as addiction mechanics and deceptive design. This is especially true when trying to persuade players to spend money and time. Psychologists, designers, and healthcare providers must work together to create games that encourage healthy interactions and build supportive communities. The success of these approaches depends on sustained, evidence-based research. Most importantly, collaboration across disciplines is crucial to create game environments that enable individual and collective thriving.

Effective game design that promotes human flourishing carefully balances player engagement with overall well-being. When game creators thoughtfully design experiences, they not only capture players' attention but also support positive psychological development. Well-designed games can foster resilience, spark

Insights

- The workshop identified eight critical research themes, including games that enable human flourishing, have an educational potential, and can prevent toxicity.
- Participants emphasized the need for collaboration between academia, industry, and policymakers.
- The workshop created a strategic framework to use games as tools for social progress, learning, well-being, and scientific advancement.

creativity, and cultivate meaningful social connections among their communities.

Realizing the potential of educational games. Educational games change the way students learn and make engaging with educational content fun and effective. These interactive experiences create more personal learning opportunities and improve educational outcomes for different subjects and student groups [2]. Despite their promise, significant barriers persist—inconsistent quality, limited curriculum integration, and insufficient teacher training—undermining the widespread adoption of games. Educational institutions and policymakers must create strong support systems. This goal can be achieved through targeted teacher development programs, robust technological infrastructure, and rigorous evaluation standards. International collaboration amplifies these efforts and makes high-quality educational gaming accessible globally. This international setup is an incubator for more adaptive, inclusive learning environments.

We deliberated on compelling evidence of educational gaming's transformative impact. Game-based math programs show clear increases in student engagement and understanding compared to traditional teaching methods. These success stories represent just the beginning of the potential of educational games. Game-based learning can help solve ongoing educational problems once it receives more resources and support from institutions. Conventional approaches have failed to solve these problems thus far. The workshop participants explored how games can play an essential role as next-generation learning spaces, responding dynamically to individual student needs. They also support collaborative problem-solving skills, which are essential for students' success.

Shifting the toxicity paradigm from mitigation to prevention. Toxic behaviors plague gaming communities and undermine player experiences and community health [3]. Proactive prevention rather than reactive measures is the solution. One way to

do this is by creating comprehensive systems that reshape community dynamics. Real-time moderation technologies, clear community standards, and player empowerment initiatives form the foundation of this approach.

Game developers now integrate sophisticated preventive mechanisms into their platforms. AI-driven moderation systems identify problematic patterns before they escalate. Positive reinforcement mechanisms reward constructive participation, and strategic community-building cultivates environments where toxicity struggles to take root. Educational initiatives centered on digital citizenship and empathy development complement these technical solutions. These measures provide multilayered protection against harmful behaviors.

Workshop participants discussed compelling evidence that supports prevention-focused approaches. When gaming communities use member-led moderation, target reward systems, and clear articulation of behavioral guidelines, dramatic reductions in harassment and toxicity happen. Thus, early intervention strategies, combined with positive reinforcement mechanisms, improve community

health. We can get frameworks not only for gaming environments but potentially all digital social spaces from implementations of this type. This direction will allow us to establish new standards for online interaction where toxicity will become the exception rather than the norm.

Games as large-scale behavioral research platforms. Games provide unparalleled environments for behavioral research at massive scale [4]. Within these virtual spaces, researchers observe human interactions, decision making, and social dynamics in real time under controlled conditions. Social phenomena unfold naturally in such environments. Players lay bare their cognitive processes and group behaviors that traditional research methods can't easily capture.

Standardized data collection frameworks transform these gaming environments into powerful research tools. Universal modding systems and procedural content generation expand research possibilities exponentially. For this potential to become reality, policymakers and industry leaders must establish ethical data-sharing protocols. The goal of such protocols is to facilitate collaboration between



Attendees of the 2025 Dagstuhl Perspectives Workshop: The Future of Games in Society.

commercial developers and academic researchers.

The workshop identified specific research applications with disruptive potential. Games are vehicles for the study of collective problem-solving at an unprecedented scale. They show social network formation patterns and shed light on decision-making mechanisms across scenarios. When we combine them with advanced artificial intelligence and machine learning techniques, these platforms can drive breakthroughs in public health, education, social policy, and AI development.

Building sustainable and scalable research infrastructure.

A robust, sustainable, and scalable infrastructure is fundamental for advancing games research. Shared research platforms, open databases, and standardized tools will kick-start teamwork. Access to such resources has the potential to spark more interdisciplinary research and innovation. Creating universal APIs, using modular designs, and providing open access resources makes research more scalable and accessible. Centralized data repositories and curated research standards cut duplication and speed up collective knowledge development.

Policymakers and funding groups should focus on ongoing investment in infrastructure and career development. A skilled and diverse workforce is the direct result of such a systematic focus. Building a strong infrastructure will keep things running smoothly. Research groups can then make use of that infrastructure to make meaningful contributions, accelerating the pace of benefits provided by games for society.

The workshop clearly showed that scalable infrastructure forms the backbone of advanced games research. Shared platforms and standardized tools have proved to be able to fast-track research progress and expand collaborative opportunities across disciplines. Establishing this critical infrastructure demands sustained institutional commitment and dedicated funding, which will improve research efficiency and social impact. It fundamentally changes how games can benefit society through research.

Bridging industry and academia to build sustainable research partnerships. Building strong partnerships between research institutions and game companies is paramount because it sparks new ideas and shows that research works in real life. Collaborative efforts support the sharing and creation of new knowledge. They give industry access to the latest research insights. At the same time, academia gains valuable perspectives on real-world challenges. To keep these partnerships strong, it's essential to address intellectual property issues first. Throughout such collaborations, aligning goals and creating clear communication channels also matter. Initiatives such as joint funding programs, industry advisory boards, and knowledge-exchange platforms keep collaborations going.

Conferences and knowledge-sharing events connect theory and practice. They create a space for growth and innovation that is mutually beneficial. In addition, policies that are clear and promote transparency protect intellectual property, which encourages participation and results in strong partnerships. This can lead to innovations that are both effective and commercially successful. Connecting those two cornerstones benefits society.

Successful industry-academic partnerships need clear rules from the start. These rules should cover data sharing, intellectual property rights, and publication policies. Workshop participants talked about how strong partnerships give rise to better game design ideas and enhance player experience research. Shared success stories illustrate how collaboration

between academia and industry speeds up research and puts findings into action right away.

Developing rigorous research standards and quality assessment methods for games research. To maximize the credibility and impact of games research, we must set strong research standards and quality assessment frameworks. Games research has been facing this challenge for years, and its interdisciplinary nature needs clear and accepted evaluation metrics. Additionally, our research methods must reflect the diverse contributions in the field. Open science practices, data transparency, and study preregistration strengthen research reliability and replicability.

To collect, share, and analyze data, participants discussed how to create centralized research platforms. These would make it easier to conduct thorough evaluations and meta-analyses. Funding replication and longitudinal studies contribute to the robustness and lasting impact of research findings. Efforts made in education and public outreach should support these standards so that more people recognize and appreciate quality research. Robust games research, when implemented effectively, can provide valuable direction for policymakers, educators, and industry professionals.

The workshop participants emphasized that standardized research methodologies and quality metrics are essential for advancing games research. Some institutions have already successfully implemented these standards, which shows that they are effective. Modern frameworks evaluate research impact through many dimensions, including citation metrics, industry adoption rates, and measurable societal benefits. As a result, games research has gained increased credibility in both academic and industry circles.

Building a global funding infrastructure for games research. Sustainable global funding is vital for advancing games research, which is the driver behind significant social change. Creating international funding programs, partnering with private companies, and setting up

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long-term endowments would provide critical resources for research. A major step that's needed is the establishment of collaborative international consortia to pool expertise and funding to tackle global challenges. If policymakers and funding agencies see how vital games research is, they will be able to support various funding models in better ways. This includes industry coinvestment and working together internationally.

Global forums and advocacy groups can support games research to show its value to society and gain public and institutional backing. Fair resource allocation and tackling research gaps in different areas will accelerate inclusive growth and innovation. This robust funding system would be able to support ongoing innovation and amplify games' role in supporting the growth of society. The new U.K. Smart Data Donation Service supports large-scale data donation in games for ongoing research [5].

Participants stressed the need to diversify funding sources. They suggested new ideas, such as research-industry matching funds and international grant programs. They also made clear that sustainable funding needs to show a clear social impact and a good return on investment. Successful funding initiatives, such as cross-border research consortia and public-private partnerships, serve as models for future developments.

FROM VISION TO ACTION

To solve complex problems, we need strong partnerships, which should include various sectors and foster collaboration. Specifically, we stress the importance of uniting universities, companies, organizations, and government agencies. Each of these has unique strengths that, when combined, create a synergistic effect. Universities provide research skills and innovation. Companies offer practical applications and market insight; organizations add community knowledge and advocacy; government agencies supply policy frameworks and resources. Such partnerships create outcomes that are far greater than

what any single entity could achieve alone. Therefore, building and keeping these partnerships across sectors is essential and helps drive real progress and create lasting positive change.

To succeed, we need detailed action plans, which should have specific timelines and clear accountability measures. The workshop also brought into focus the importance of community engagement. Player feedback directly influences research and development priorities. In addition, regular meetings with policymakers are vital in order to ensure that public policy is well-informed.

THE PATH FORWARD

The workshop united different viewpoints and created practical strategies for the future while not compromising on the ultimate vision of what games can do for society at large. This has led to a clear plan for using games to drive positive change, which we are currently drafting into a manifesto for the future of games in society. Researchers, developers, and other stakeholders can now collaborate to shape this new landscape (exemplified by the U.K. government's Video Games Research Framework) [6]. We need to focus on creating meaningful experiences that support wider societal goals.

The Dagstuhl seminar showed how combining academic research, industry innovation, and policy development could reinvent games research. We worked with experts across disciplines to create frameworks for addressing societal challenges through gaming. We believe that this interdisciplinary approach lets us create evidence-based solutions that are both inclusive and ethically sound. If we keep working together, games can change from just being a fun pastime to becoming effective tools for positive social change.

With games seamlessly woven into the fabric of modern society, the vision articulated at Dagstuhl presents a compelling case for games as powerful tools for human connection, learning, well-being, and

social progress. The challenge lies in translating this vision into reality, a task that will require sustained collaboration across disciplines, sectors, and borders.

Authors' note: Claude 3.7 Sonnet was used in generating the initial draft of this article.

ENDNOTES

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